

Column Selection Report (EDA Phase)

Objective

The goal of this step was to identify which columns should be retained or dropped for Exploratory Data Analysis (EDA), based on: - Data coverage - Semantic relevance - Analytical usefulness - Redundancy and noise

The intent was not to delete data, but to construct a clean, interpretable EDA dataset that supports reliable insights.

Columns Dropped (and Reasons)

The following columns were excluded from EDA due to high missingness, redundancy, inconsistency, or limited analytical value **for this phase**.

Review Breakdown & Community Metrics

Column	Reason for Dropping
positive_reviews	Highly correlated with n_reviews; absolute counts add limited value without normalization
negative_reviews	Redundant with positive_reviews and n_reviews
positive_ratio	Derived metric; can be recomputed later if required
avg_helpful_votes	Influenced by exposure bias; noisy for first-pass EDA

Conclusion: These metrics are either derived or redundant and are deferred to advanced sentiment modeling stages.

System Requirement (Recommended) Columns

Column	Reason for Dropping
mat_pc_graphics_rec	>60% missing; free-text and non-standardized
mat_pc_memory_rec	Same as above
mat_pc_processor_rec	Same as above
mat_pc_os_rec	Low variance and incomplete coverage

3 Achievement Metadata

Column	Reason for Dropping
mat_achievement_count	Missing for most apps; weak explanatory power

4 Text, URLs, and Presentation Fields

Column Type	Reason for Dropping
Descriptions (about_the_game, short_description)	Text analysis out of EDA scope
Images / URLs (header_image, background)	Non-analytical
Language lists	High cardinality and noisy
Operational timestamps (created_at, updated_at)	Do not explain user behavior

Columns Retained (and Reasons)

The following columns were explicitly retained because they are interpretable, well-covered, and analytically meaningful.

1 Identifiers

Column	Reason
appid	Unique identifier; join key
name	Human-readable reference

2 App Metadata

Column	Reason
type	Distinguishes games, DLCs, and software
is_free	Core business model variable
required_age	Content/access constraint
release_date	Enables temporal analysis

3 Pricing & Monetization

Column	Reason
mat_initial_price	Original pricing anchor
mat_final_price	Effective consumer price
mat_discount_percent	Promotional behavior

4 External & Popularity Signals

Column	Reason
metacritic_score	External critical reception signal for high-visibility titles
recommendations_total	Proxy for community endorsement and reach

5 Content Descriptors

Column	Reason
genres	Strongest explanatory variable
categories	Gameplay features
developers	Production patterns
publishers	Distribution & scale effects

6 Engagement

Column	Reason
avg_playtime	Measures engagement depth

7 Platform Support

Column	Reason
mat_supports_windows	Market reach
mat_supports_linux	Niche adoption

Column	Reason
mat_supports_mac	Platform bias

Summary of Changes

- **Moved to Dropped:** positive_reviews, negative_reviews, positive_ratio, avg_helpful_votes
- **Moved to Retained:** metacritic_score, recommendations_total

This revision aligns the EDA dataset toward **core popularity, pricing, content, and platform signals**, while deferring detailed sentiment decomposition to later analysis stages.